



PAK-AUSTRIA FACHHOCHSCHULE:
INSTITUTE OF APPLIED SCIENCES AND TECHNOLOGY

ICT Lab Report #10

Course Information

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Introduction

This laboratory report documents the completion of two essential web development tasks using HTML. The tasks focus on creating semantic HTML structures for real-world web applications. Task 1 involves creating a professional CV (Curriculum Vitae) webpage that showcases personal information, skills, work experience, and education in a structured format. Task 2 involves building a student registration form that collects user information through various HTML input elements. Both tasks demonstrate proper use of HTML semantic tags and form elements to create functional and well-structured web pages.

Task 1: CV Creation - Professional Webpage

Objective

The objective of this task is to create a professional CV webpage using HTML semantic tags. This webpage should effectively present personal information, professional skills, work experience, and educational background in an organized and visually logical manner. The CV webpage will serve as a digital portfolio that can be shared with potential employers or clients.

Task Description

In this task, students create a complete CV webpage with the following components:

- Navigation bar for easy section access
- Header with personal information
- About Me section describing professional background
- Skills section listing competencies
- Work experience section detailing past employment
- Education section displayed in a table format
- Footer with contact information

Step-by-Step Implementation

Step 1: Create and Save the HTML File

First, open your code editor (such as Cursor AI, Visual Studio Code, or any HTML editor) and create a new file. Save this file with the filename **cv.html** in your working directory. This file will contain the complete CV webpage code.

Key Point: The .html extension tells the browser that this is an HTML document and should be rendered as a webpage.

Step 2: Write the Basic HTML Structure

Every HTML document requires a standard structure. This structure consists of essential tags that form the foundation of any webpage:

CV Webpage

Explanation of each component:

HTML Tag	Purpose
<!DOCTYPE html>	Declares the document type as HTML5 to the browser
<html>	Root element that wraps all content
<head>	Contains metadata like page title and character encoding
<title>	Sets the browser tab title
<body>	Contains all visible content displayed on the webpage

Step 3: Add the Navigation Bar

The navigation bar is placed at the top of the page and provides links to different sections of the CV. Using the <nav> tag with <a> (anchor) tags creates an organized navigation structure:

About

|

[Skills](#)

|

[Experience](#)

|

[Education](#)

|

[Contact](#)

Why this matters:

- The href attribute uses the # symbol to link to section IDs on the same page
- This allows users to jump directly to the desired section
- Improves user experience and navigation efficiency
- Uses semantic HTML that search engines understand

Step 4: Create the Header Section

The header displays the applicant's name and basic contact information:

Eshal Ayub

Email: Eayub006@gmail.com | Phone: 0332-9466454 | Location: Abbottabad

Component explanation:

- <header> tag semantically indicates this is introductory content
- <h1> displays the person's name as the main heading
- <p> tags contain contact information in a readable format

Step 5: Add the "About Me" Section

The About Me section provides a brief personal introduction and professional summary:

About Me

I am a passionate data scientist with a strong interest in data analysis, statistics, machine learning, and problem-solving. I aim to extract meaningful insights from data and apply them to real-world problems while maintaining ethical standards and continuous learning.

Semantic significance:

- `<section>` groups related content together
- `id="about"` allows the navigation bar to link to this section
- `<article>` indicates this is self-contained, standalone content
- This structure is SEO-friendly and accessible

Step 6: Create the Skills List

Skills are presented using an unordered list, which is the standard way to display lists in HTML:

Skills

- Data Analysis
- User Research
- Usability Testing
- Project Management

Why use `<ul>` and `<li>`:

- `<ul>` (unordered list) displays items with bullet points
- `<li>` (list item) represents each individual skill
- This format is easier to read than plain text
- Screen readers can interpret lists properly for accessibility

Step 7: Add the Work Experience Section

Work experience is structured with multiple `<article>` tags, each representing a different job:

Work Experience

Senior Data Architect

Proseware, Inc. (Jan 2022 – 2023)

Managed design teams, created wireframes and prototypes, and collaborated with clients to deliver effective solutions.

Data Analyst

Future Prep (Oct 2021 – 2022)

Led redesign projects, conducted user research, and improved system performance and user experience.

Structure benefits:

- Each job is in its own `<article>`, making the HTML structure clear
- Job title uses `<h3>` for proper heading hierarchy
- Dates and descriptions are easy to locate and modify
- This semantic approach is crawler-friendly

Step 8: Create the Education Section Using a Table

Education details are best displayed in a table format for clarity and organization:

Education

Degree	Institution	Year
BFA (Graphic Design)	Pak-Austria	20XX

Table components:

- `<table>` creates the table structure
- `<tr>` (table row) creates horizontal rows
- `<th>` (table header) creates column headers with bold text
- `<td>` (table data) contains the actual data cells
- `border="1"` adds visible borders between cells

Step 9: Add the Footer Section

The footer is placed at the bottom and contains additional contact or reference information:

References available upon request

Purpose:

- Provides closing information
- The `id="contact"` allows the navigation to link here
- Completes the webpage layout logically

Step 10: Save and Run the File

After writing all the code:

1. **Save the file** as `cv.html` on your computer
2. **Open the file** in any web browser (Chrome, Firefox, Edge, Safari)
3. **Verify functionality:**
 - All sections are visible and properly formatted
 - Navigation links work correctly
 - Table displays properly
 - Contact information is readable

Key Concepts Demonstrated

Concept	Explanation
Semantic HTML	Using HTML tags that describe meaning to both browsers and developers
Navigation	Creating links within the same page using anchors and IDs
Content Organization	Grouping related content using sections and articles
Tables	Organizing tabular data in an easily readable format
Accessibility	Creating structure that can be interpreted by screen readers

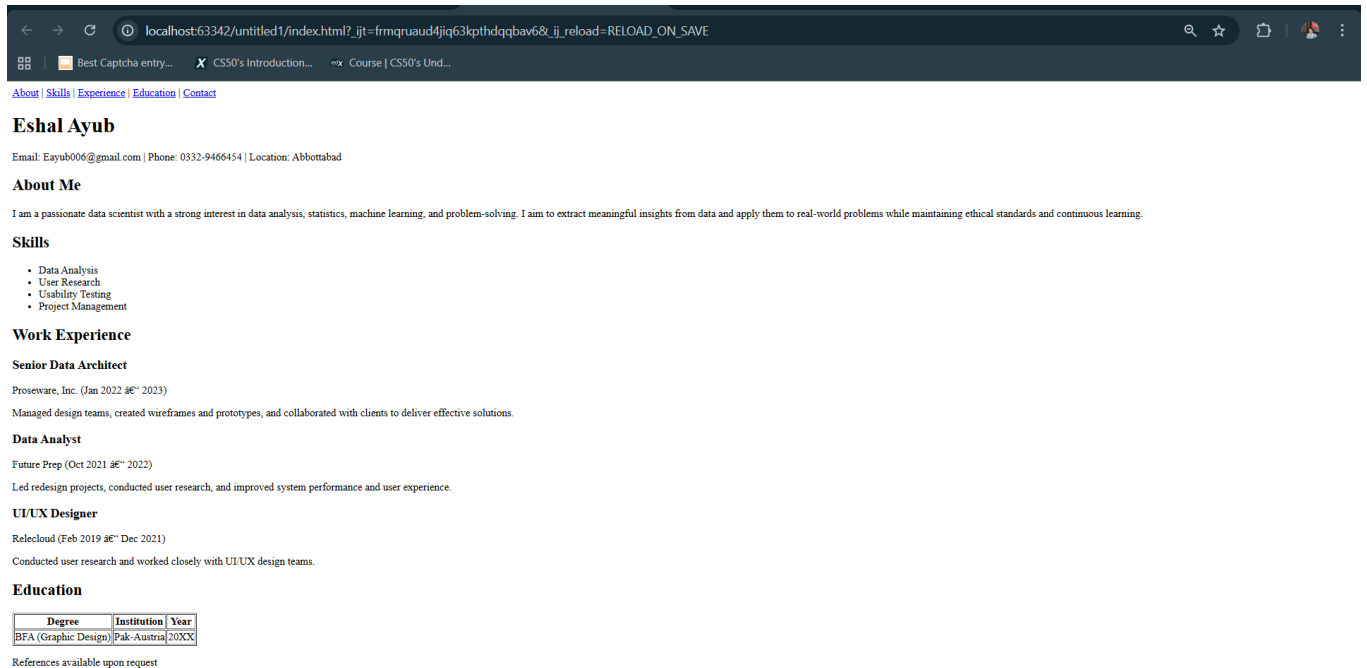
Table 1: Key HTML Concepts in Task 1

Conclusion for Task 1

Task 1 successfully demonstrates the creation of a professional CV webpage using only HTML semantic tags. The page showcases:

- Proper HTML document structure with DOCTYPE, html, head, and body tags
- Semantic tags (header, section, article, footer, nav) for proper content organization
- Navigation functionality using anchor links
- Professional presentation of personal information
- Organized display of skills, experience, and education
- Clean and readable layout that follows web standards

This webpage can be further enhanced with CSS styling to improve visual appearance and responsiveness on different devices.



http://localhost:63342/untitled1/index.html?_ijt=vlcrbiv6i6fvb86s67mjguqtk4&_ij_reload=RELOAD_ON_SAVE

Task 2: Student Registration Form Creation

Objective

The objective of this task is to create an interactive student registration form using HTML form elements. This form collects personal and educational information from students in a structured and organized manner. Students will learn how to use various HTML input elements such as text fields, email inputs, radio buttons, checkboxes, dropdown menus, and submit buttons.

Task Description

In this task, students create a complete registration form webpage with:

- Navigation bar linking to different parts of the form

- Form heading explaining the purpose
- Personal Details section (name, email, gender, phone, city)
- Education Details section (program, skills)
- Instructions table explaining form requirements
- Submit button to process the form
- Footer with copyright information

Step-by-Step Implementation

Step 1: Create and Save the HTML File

Open your code editor and create a new file. Save this file as **form.html** in your working directory. This file will contain the complete student registration form.

Step 2: Write the Basic HTML Structure

Start with the essential HTML structure:

Student Registration Form

This basic structure provides the foundation for the registration form webpage.

Step 3: Add the Navigation Bar

Create a navigation bar at the top of the form:

[Home](#)

|

[Register](#)

|

[Contact](#)

Navigation purpose:

- Provides links to different sections of the website
- Improves user navigation and website structure
- # is used as placeholder for actual page links
- The pipe character | separates the navigation items visually

Step 4: Add the Form Heading and Instructions

Add a heading and create an instructions table:

Student Registration Form

Instruction	Details
Fill all fields	Mandatory

Important note: The instructions table communicates important information to users about form requirements.

Step 5: Create the Form Element

All form inputs must be wrapped inside the `<form>` tag:

The `<form>` tag is essential because:

- It defines the area where user input is collected
- It prepares data to be submitted to a server
- All input elements must be contained within this tag

Step 6: Create the Personal Details Section

Add a section for collecting personal information:

Personal Details

Name:

Email:

Gender:

Male

Female

Contact Number:

City:

Input element explanations:

Input Type	Purpose and Example
type="text"	Single-line text input for name, city, phone numbers
type="email"	Email input with built-in validation for email format
type="radio"	Radio buttons for single selection (e.g., choosing one gender)
value attribute	Pre-fills the input field with a default value
checked attribute	Pre-selects a radio button or checkbox by default
 tag	Creates line breaks for better spacing and readability

Table 2: HTML Form Input Types

Step 7: Create the Education Details Section

Add a section for education information:

Education Details

Program:

Skills:

Data Analysis

User Research

HTML

CSS

Dropdown and checkbox explanation:

- `<select>` creates a dropdown menu for program selection
- `<option>` defines each choice in the dropdown
- `selected` attribute pre-selects an option
- `<input type="checkbox">` allows multiple selections
- Checkboxes are useful when a user can select multiple items

Step 8: Add the Submit Button

Create a submit button to process the form:

Submit button function:

- The `type="submit"` makes it a clickable button
- When clicked, the form data would be sent to a server
- The `value` attribute displays the text on the button

Step 9: Add the Footer Section

Complete the form with a footer:

© 2025 Student Registration System

Footer purpose:

- Provides copyright and footer information
- Completes the webpage layout
- Offers professional appearance

Step 10: Save and Run the File

After completing the code:

1. **Save the file** as `form.html`
2. **Open in a web browser** to view the form
3. **Test the form elements:**
 - Type in text fields
 - Select radio buttons
 - Choose from dropdown menu
 - Check/uncheck checkboxes
 - Verify the submit button appears

Form Elements Summary

Element	HTML Code	Function
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Text Input	<code><input type="text"></code>	Collects text information
Email Input	<code><input type="email"></code>	Collects email with validation
Radio Button	<code><input type="radio"></code>	Single selection from options
Checkbox	<code><input type="checkbox"></code>	Multiple selections possible
Dropdown	<code><select></code>	Menu-based selection
Submit Button	<code><input type="submit"></code>	Submits form data

Table 3: HTML Form Elements Reference

Conclusion for Task 2

Task 2 successfully demonstrates the creation of an interactive student registration form using HTML form elements. The form includes:

- Proper HTML document structure
- Organized sections for different types of information
- Various input types for different data collection needs
- A table to display instructions
- Professional layout with proper spacing
- All form elements contained within the `<form>` tag

The form collects essential student information in an organized and user-friendly manner. Future enhancements could include CSS styling and JavaScript validation for improved user experience.

localhost:63342/untitled2/index.html?_ijt=vlcrbiv6i6f6vb86s67mjguqtk4&_ij_reload=RELOAD_ON_SAVE

Home | Register | Contact

Student Registration Form

Instruction Details

Fill all fields Mandatory

Personal Details

Name:

Email:

Gender: ☐ Male ☒ Female

Contact Number:

City:

Education Details

Program:

Skills:

- ☒ Data Analysis
- ☒ User Research
- ☐ HTML
- ☐ CSS

© 2025 Student Registration System

http://localhost:63342/untitled2/index.html?_ijt=vlcrbiv6i6f6vb86s67mjguqtk4&_ij_reload=RELOAD_ON_SAVE

Summary and Key Learning Outcomes

Through these two tasks, students have learned:

1. **Semantic HTML:** Using appropriate HTML tags that describe content meaning
2. **Page Structure:** Creating well-organized websites with headers, sections, and footers
3. **Navigation:** Implementing internal links within webpages
4. **Forms:** Collecting user input through various form elements
5. **Tables:** Organizing tabular data in a structured format
6. **Lists:** Presenting information as bulleted or numbered lists
7. **Accessibility:** Creating HTML that is interpretable by both browsers and assistive technologies
8. **Best Practices:** Following HTML standards and conventions for professional web development

Recommendations for Further Enhancement

1. **CSS Styling:** Add CSS to improve visual appearance with colors, fonts, and spacing
 2. **Responsive Design:** Use media queries to make forms work on mobile devices
 3. **JavaScript Validation:** Add form validation to check data before submission
 4. **Backend Integration:** Connect forms to server-side processing
 5. **Accessibility Features:** Add ARIA labels and improve keyboard navigation
 6. **Mobile Optimization:** Ensure forms work properly on all screen sizes
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